

HW 1

Linear Regression

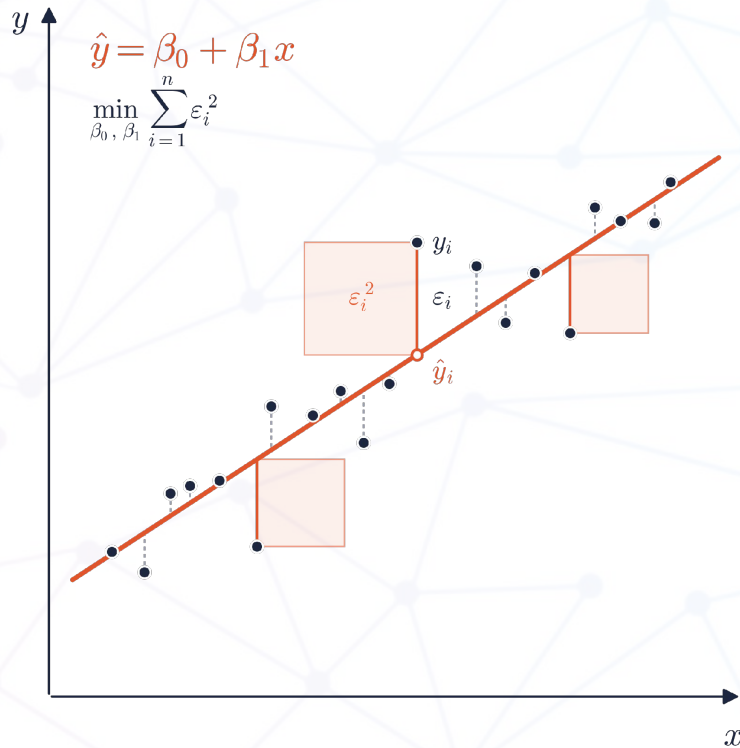
Assignment specification and grading criteria

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Submission Deadline Mon 2026-10-19 23:59

Presentation Tue 2026-10-20, in class



OVERVIEW

Task Introduction

Each team defines its own prediction application built on **linear regression**. The task starts from a **real need**, **targets a continuous quantity that is worth predicting**, and produces a result that can be evaluated.

Permitted subjects are unrestricted. Representative examples:

House price / rent

Store foot traffic

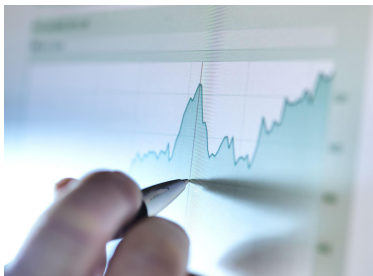
Electricity demand

Used-car valuation

Crop yield

Project effort hours

Model complexity is not assessed. Assessment covers the clarity of the problem definition, the deliberateness of the feature design, and the verifiability of the result. Creative, distinctive projects also earn higher scores.



Scope of the Assignment

The prediction task is defined by the team. The data, the features, the method, and the evaluation setup are documented in the report.



You decide

- Application scenario and data source
- Target variable to predict
- Feature design and data representation
- Linear model variant: OLS, Ridge, Lasso, Elastic Net, or polynomial / interaction expansion
- Which AI tools to use, and for what



Fixed for every group

- The final predictor is a linear regression model; other model families may appear only as baselines
- The target is a continuous quantity with a clear unit
- Real-world data with a documented source (public or self-collected)
- All methods share the same data split and metrics
- Findings are reported as a short ACL-style paper

Paper Report: Write a Short ACL Format Paper

Present your work as a research paper, not as answers to a question list. The next slides guide each section.

(EXAMPLE)



Paper report example in the official ACL template.



Template

Official ACL LaTeX template on Overleaf: [open template](#)
Use `\usepackage[preprint]{acl}` (shows names and page numbers).



Length

At most 6 pages, excluding references. Content after page 6 is not read. Keep the template's margins and font size.



Front matter

Title, team ID, every member's name and student ID, and the URL of the project's GitHub repository.



Figures & tables

Numbered, captioned, and referenced in the text; axes labeled with units.

Paper Structure and Section Weights

Each section answers one guiding question. *The paper report is scored out of 100 as below, then weighted 70% of HW1.*

Section	Guiding question	Points
Abstract	Why does this prediction matter, to whom, and what do you contribute?	7
1 Introduction		
2 Related Work	What have others done, and how does your work differ?	6
3 Task Definition & Data	What exactly is predicted, from which data, prepared how?	12
4 Exploratory Feature Analysis	What does the data suggest before any modeling?	10
5 Method	How do raw data become a prediction, and why this design?	13
6 Experimental Setup	How are models validated, measured, and compared?	10
7 Results	Does your design beat the baselines, and do your features help?	18
8 Analysis	What did the model learn, and where does it fail?	16
9 Conclusion & Limitations	What should the reader take away, and what remains open?	5
Contributions of Each Member	Who did what, and how were AI tools used?	3
References	Real, verifiable sources in ACL citation style (outside the 6 page limit)	—
Paper Report Total		100

Use these sections, in this order, as the headings of your paper; section titles may be adapted to your topic.

Abstract & Introduction

Motivate the prediction problem and state what your paper contributes.

Write about

- The application and why the prediction matters
- The target users and the decision the prediction supports
- A precise research question that the paper answers
- Two or three contribution statements, each verified later in §7–§8
- **Abstract (≤ 150 words):** task, approach, and the main quantitative result

(EXAMPLE)

Weak motivation: “We predict solar output.”

Strong motivation: “A building manager decides every hour whether to charge the battery or buy grid power; our forecast feeds that decision.”

Research question: “Can physically motivated features let a linear model forecast next-hour PV output well enough for battery scheduling?”

(EXAMPLE)

Running example: a 50 kW rooftop solar plant

Input features

Next-hour weather forecast and last measured output



Linear model

Ridge regression



Predicted value

PV output in the next hour (kWh)



User decision

Battery dispatch and grid purchase plan

All **EXAMPLE** boxes on the following slides **show the expected form only**.

Common deductions Vague motivation with no identified user or decision; contributions that later results do not support.

Related Work

Position your work against existing studies, datasets, and deployed systems.

Write about

- Cite at least 3 relevant sources: papers, technical reports, benchmark datasets, or deployed systems
- For each, summarize the task, data, features, model, and reported error
- State explicitly what your work does differently: a new feature, setting, dataset, or user
- Cite with BibTeX and `\citet{}` / `\citep{}` as in the ACL template



Every reference must be real and verifiable. Check each citation on the conference proceedings or the publisher's page. **Fabricated references are penalized.**

(EXAMPLE)

A compact comparison table

Work	Data	Features	Model	nRMSE
Prior [1]	1 utility plant, 1 yr	Measured irradiance, temperature	OLS	9.8%
Prior [2]	3 rooftop sites, 2 yr	Weather forecasts	Gradient boosting	6.5%
Ours	50 kW rooftop, 2 yr	Forecasts + engineered features	Ridge	3.3%

All three predict hour-ahead PV output. nRMSE = RMSE / plant capacity; errors on different data give context, not a direct comparison. [1], [2] stand for cited papers.

Common deductions Citations listed without any comparison; unverifiable or fabricated references.

Task Definition & Data

Define the prediction task precisely, then describe how the data were obtained and prepared.

3.1 Task definition (5%)

- **Target:** quantity, unit, and plausible range
- **Unit of prediction:** what one row represents
- **Inputs and prediction time*:** which features are known when the prediction is made

***Rule:** Use only information available at prediction time; using future values for prediction is not allowed.

3.2 Data (7%)

- Source, collection method, period, and size, with a **table of summary statistics**
- **Preprocessing and the reason for each step** (missing values, outliers, scaling, encoding)

(EXAMPLE)

Task card for the running example

Item	Definition
Target	PV output in the next hour (kWh); range 0–50
One row	One plant-hour between 06:00 and 18:00
Inputs	Forecast irradiance, temperature, humidity, cloud cover, wind; time of day; last measured output
Known at	One hour ahead: forecasts and past measurements only
Data	2024–25 hourly meter log + forecast API; 9.5k rows

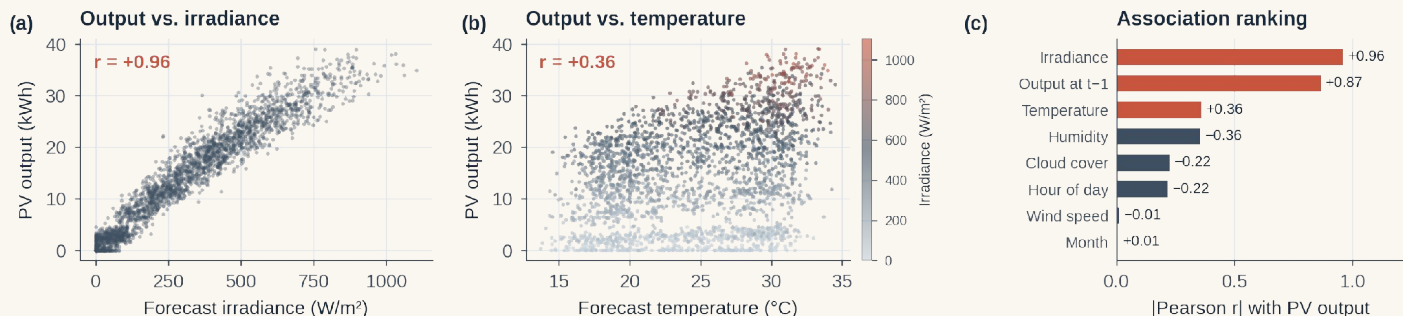
Common deductions Undefined unit or range; inputs that are unavailable at prediction time; preprocessing without justification.

Exploratory Feature Analysis

Show the evidence that led to your features, before building any model.

- At least one figure of the target against key candidate features (scatter, box, or line plots)
- Quantify association: Pearson / Spearman r or mutual information; check redundancy among features
- Explain how this evidence led you to keep, transform, or drop features; state hypotheses (H1, H2, ...) to test in §8.1
- Use the training portion only, so the test set remains unseen

(EXAMPLE)



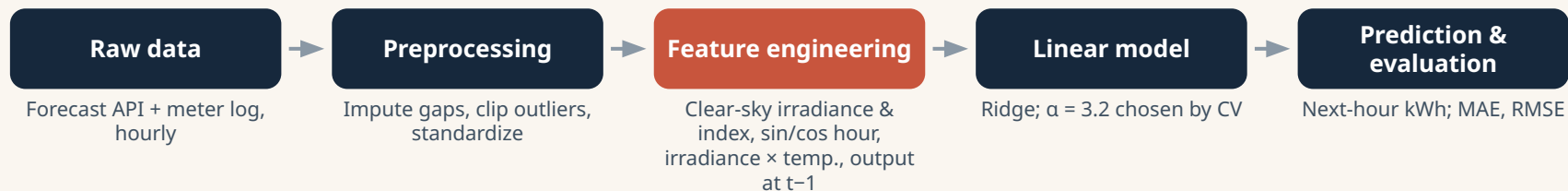
H1: output rises almost linearly with irradiance (a). H2: temperature's positive r may only reflect that hot hours are sunny hours (b). The top of the ranking in (c) forms Set B in §7.2 (training split only).

Common deductions Figures without interpretation; statistics computed on test data; hypotheses that are never revisited.

Pipeline & Feature Engineering

Describe how raw data become a prediction. A pipeline figure showing each stage's input and output is required.

(EXAMPLE)



Design one or more feature groups beyond the raw columns, each motivated by §4 or by domain knowledge:

Transforms & interactions

log, square, and product terms; let a linear model capture curvature and joint effects
e.g., $\log(\text{income})$; $\text{floor area} \times \text{district}$

Aggregates

lags, rolling means, group averages; add temporal or group context
e.g., sales over the previous 7 days

Domain-derived variables

quantities an expert would compute; encode known physics or rules
e.g., body-mass index from height and weight

For each group, state in one or two sentences why it should help and which hypothesis from §4 it addresses.

Common deductions Missing pipeline figure; features added without a stated rationale.

Model Choice & Hyperparameter Tuning

Justify which linear model you use, and show how its hyperparameters were chosen.

Requirements

- Name the linear variant and justify it with properties of your data, such as collinearity found in §4, not by trial and error alone
- Tune α , the L1 ratio, or the polynomial degree by cross-validation on the training portion only (§6)
- Report the search range and the selected values

Choosing a variant (example)

Variant	Penalty	Typical reason to choose it
OLS	none	Few features, little collinearity
Ridge	L2	Correlated features; shrinks, keeps all
Lasso	L1	Many weak features; sets some to zero
Elastic Net	L1 + L2	Correlated groups plus selection
Polynomial terms	any	Curvature, joint effects; linear in β

(EXAMPLE)

Model paragraph for the running example, as it would appear in the report

5.2 Model

Our features are strongly correlated (VIF up to 40, §8.1), so we use ridge regression, which keeps every feature but shrinks unstable coefficients. Features are standardized inside a scikit-learn Pipeline. We select α from 13 log-spaced values in $[10^{-3}, 10^3]$ by 5-fold TimeSeriesSplit on the training period only, minimizing MAE; the selected value is $\alpha = 3.2$. Because this α is small relative to $n = 7,514$, the ridge coefficients differ from OLS by less than 0.05, so §8.1 reports OLS estimates with HAC errors.

α	10^{-3}	10^{-2}	10^{-1}	1	3.2	10	10^2	10^3
CV MAE	1.303	1.303	1.303	1.303	1.303	1.304	1.344	1.817

Tuning curve (kWh): CV MAE is flat for $\alpha \leq 10$, so the exact choice matters little there; larger α over-shrinks and error rises.

Common deductions Model choice without a stated reason; hyperparameters tuned on the test set or not reported.

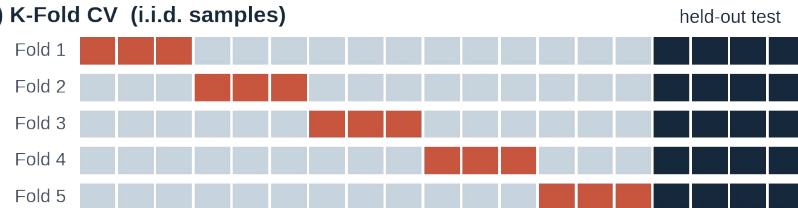
Validation Protocol

Estimate how well the model generalizes to unseen data, not how well it fits the training data.

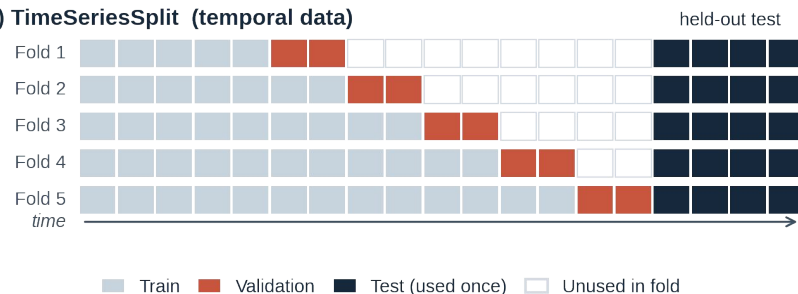
- Hold out a final test set; evaluate on it once, after all design decisions are fixed
- Within the training portion, use **K-Fold CV** for i.i.d. samples, **or TimeSeriesSplit** for temporal data, where every validation fold follows its training data; shuffling time-ordered rows lets the model learn from the future
- Use **GroupKFold** when one entity contributes many rows
- Report mean $\pm$ std across folds and justify the number of folds
- Use identical folds for every method (fixed indices or random seed)

Workflow split off the test set $\rightarrow$ tune and compare with CV on the training portion $\rightarrow$ refit the chosen model on all training data $\rightarrow$ evaluate once on the test set

(a) K-Fold CV (i.i.d. samples)



(b) TimeSeriesSplit (temporal data)



Common deductions Shuffled splits on time-series data; the test set reused for tuning or model selection.

Metrics & Baselines

Choose metrics that reflect the user's cost of error, and compare against baselines of increasing strength.

Metrics (example)

Metric	Unit	Emphasizes	Caution
MAE	target	Typical error size	Weights all errors linearly
RMSE	target	Large errors	Sensitive to outliers
MAPE	%	Relative error	Unstable when $y \approx 0$
R^2	none	Variance explained	Can be negative on test data

Choose at least two metrics and justify them. State a **decision threshold** in target units: the error level at which the user's decision changes. Define it once here; §7 compares every result against it.

(EXAMPLE)

MAE ≤ 1.5 kWh (3% of capacity): below this error, the operator can skip reserve purchases.

Common deductions Metrics without justification; no decision threshold; a missing strong baseline.

Baselines



Trivial

Training-set mean; persistence (last observed value) for temporal data



Simple

Single-feature OLS on the most intuitive feature



Strong

A flexible nonlinear model (e.g., random forest, gradient boosting) on the raw inputs; losing to it is acceptable if you explain why

Main Results

Compare your full method with every baseline under the protocol defined in §6.

(EXAMPLE)

Method	CV MAE	CV RMSE	Test MAE	Test R ²
Training mean	8.85 ± 1.49	10.30 ± 1.54	7.48	-0.03
Persistence (t-1)	4.51 ± 0.78	5.22 ± 0.91	4.00	0.72
Single-feature OLS	2.18 ± 0.18	2.83 ± 0.28	2.08	0.90
Random forest (raw)	1.52 ± 0.32	2.08 ± 0.43	1.23	0.96
Ours (Ridge + features)	1.30 ± 0.17	1.67 ± 0.24	1.16	0.97

All values in kWh. CV: mean ± std over 5 TimeSeriesSplit folds. Ours vs. random forest: paired t-test over folds, $p = 0.14$; day-block bootstrap 95% CI of the test-MAE gap: [0.02, 0.11].

Experimental Results. Ours (1.16) meets the 1.5 kWh threshold; the single-feature model (2.08) does not. Against the random forest, the fold-level test is not significant (only 5 folds), but the test-set CI excludes 0: claim a small, consistent gain, not a decisive win.

Write about

- Bold the best score in each column
- Compare the results with the decision threshold from §6
- Explain why your method wins or loses against the strong baseline
- Test whether the gap to the strongest baseline is significant and name the test (e.g., paired t-test over folds, block bootstrap CI on the test set)

Common deductions Missing baselines; improvements claimed although they lie within fold-to-fold variation.

Feature Ablation

Isolate the value of your features: same model, split, and tuning; only the feature set changes.

A

Set A · Raw

All original columns

B

Set B · Top-k correlated

The k features most correlated with the target (ranking from §4), selected inside each training fold; justify k

C

Set C · Ours

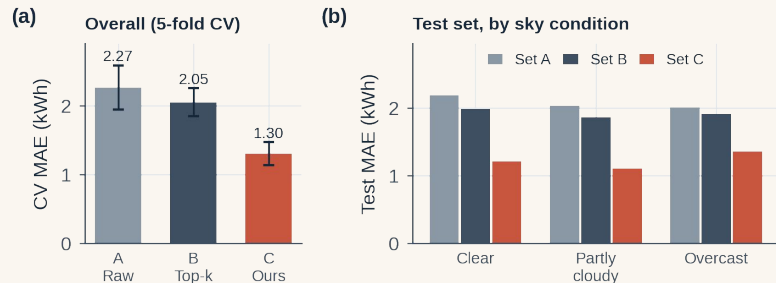
Your final set: raw columns plus engineered features, which may replace raw ones

Set B asks whether sorting correlations alone would have been enough. Optional: also remove one engineered group at a time from Set C.

Answer

- Does Set C beat Sets A and B by more than the fold-to-fold variation? If the gap is small, discuss what your engineered features add beyond a correlation ranking.
- In which data slices (e.g., overcast hours, weekends) do your features help or hurt? Report slice-level metrics
- Are the gains worth their cost: data acquisition, complexity, and interpretability?

(EXAMPLE)



Set A trails irradiance alone (2.18, §7.1): the raw month column extrapolates across time-ordered folds; without it, Set A reaches 2.17.

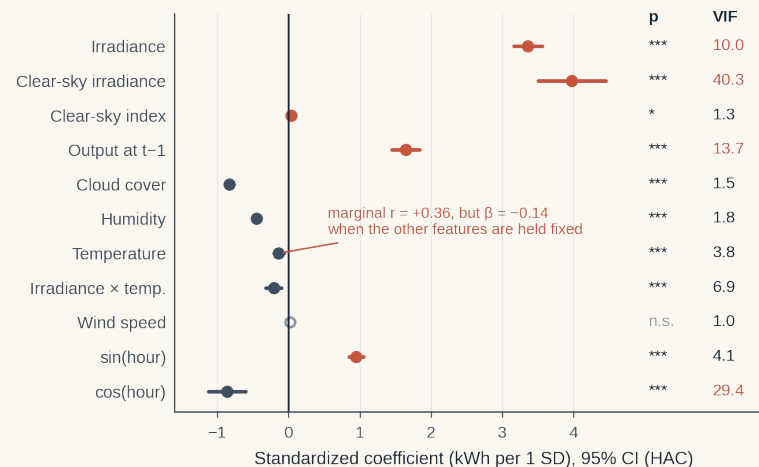
Common deductions Settings that differ in more than the feature set; top-k features selected on the full dataset.

Coefficients & Statistical Significance

Interpret what the linear model learned, and quantify how certain each effect is.

- Standardize features, then report each coefficient with a 95% CI and p-value, naming the test: t-test (OLS), bootstrap (Ridge / Lasso), HAC / Newey–West errors (autocorrelated data)
- Check each sign and magnitude against your hypotheses from §4
- Explain reversals: collinearity (report VIF), confounding, omitted variables, sampling bias

(EXAMPLE)



OLS with HAC errors on all 11 features of the final model (Ridge with $\alpha = 3.2$ gives coefficients within 0.05). Temperature: $r = +0.36$ in §4, but $\beta = -0.14$ kWh per SD with the other features held fixed, supporting H2; the negative irradiance $\times$ temp. term shows heat costs most in sunshine. Red VIF (≥ 10): collinear, interpret with care.

Minimum requirement A table or forest plot with β , 95% CI, p-value, and VIF for every feature in the final model.

Statistical $\neq$ practical significance With thousands of rows, even a negligible effect reaches $p < 0.05$. State each key effect in target units and judge whether it is large enough to matter.

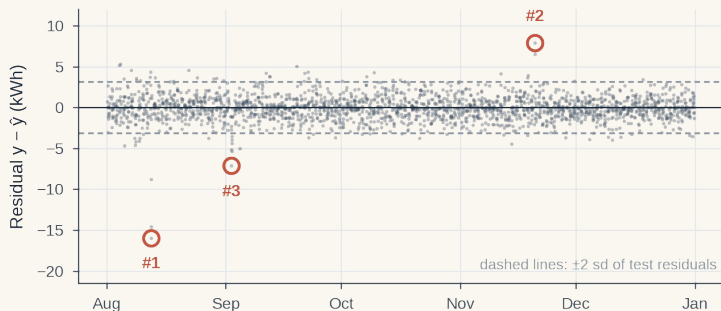
Common deductions Comparing unstandardized coefficients across units; p-values reported without naming the test.

Error Analysis

Examine where the final model fails, and why.

- Select at least 3 of the largest-error test cases; report actual, predicted, and error in target units, with the input context
- Explain each case: missing information, regime change, measurement noise, or extrapolation
- Identify a shared pattern (e.g., residuals over time or against a feature) and the first fix you would try

(EXAMPLE)



Shared pattern: all three failures come from information the model never sees (storms, soiling, fog), so the first fix is better inputs, not a bigger model.

#	Time (2025)	Actual	Pred.	Error	Likely cause
#1	08-12 14:00	7.6	23.5	-15.9	Thunderstorm missing from the forecast (cloud 45%)
#2	11-20 09:00	17.7	9.8	+7.9	Fog cleared earlier than forecast (cloud 95%)
#3	09-02 10:00	19.4	26.5	-7.1	Soiled panels after a typhoon; no plant-state feature

Values in kWh; error = actual - predicted.

Common deductions Fewer than 3 cases; cases without numbers; explanations not linked to features or data.

Conclusion & Limitations

Close the loop: answer the question posed in §1, and state honestly what the evidence cannot show.



Conclusion

- Answer the research question from §1 with your key numbers
- Summarize which features mattered, and why
- Give one actionable recommendation for the target user
- Introduce no new results



Limitations

- Data coverage: period, locations, population
- Linearity, and extrapolation beyond the observed range
- Assumptions behind your metrics and decision threshold
- The next experiment you would run

Common deductions Restating results without answering the research question; no limitations discussed.

Contributions of Each Member

List every member's concrete contributions as bullet points, and disclose how AI tools were used.

Example as it would appear in the report

Contributions of Each Member

- **Member A (student ID):** collected forecast and meter data; built the preprocessing pipeline; wrote §3.
- **Member B (student ID):** exploratory analysis and feature engineering; wrote §4–§5.
- **Member C (student ID):** validation, baselines, ablation, significance tests; wrote §6–§8.
- **All members:** presentation slides and proofreading.

AI Usage Statement

We used [tool name and version] to draft plotting code and polish wording. Feature design, experiment design, and all conclusions were decided and verified by the team.

Requirements

- One bullet per member, naming concrete tasks and the sections or figures produced
- AI usage: which tools, at which steps, and which decisions the team made and verified itself
- Place this section before the references; it counts toward the 6 pages
- Using AI tools is allowed; the team remains responsible for every result



Free-riding? Teammates normally share one score, based on overall performance on each part. **If contributions differ clearly, we may adjust individual scores accordingly.** If a teammate does not contribute, contact the TAs.

Oral Presentation

30%

of the HW1 grade

Tell the story of your paper: a clear problem, evidence-based design choices, and honest results.

What to cover

- 1 Title & team**
- 2 Motivation**
what, for whom, and why
- 3 Problem formulation**
target, inputs, timing
- 4 Data & key insight**
the plot that shaped your features
- 5 Method**
pipeline, features, model choice
- 6 Challenges**
data issues, dead ends, surprises
- 7 Evaluation**
protocol, threshold, baselines, ablation
- 8 Interpretation**
key coefficients, failure cases
- 9 Related work**
how your approach differs
- 10 Contributions**
who did what
- 11 Conclusion**
takeaways for the user



Date

Tue 2026-10-20, in class



Slides

PPTX or PDF, submitted with the report by 2026-10-19; present the submitted file



Time limit

Strictly 8 minutes per group



Graded on

Clarity of the story, evidence behind claims, readable figures, timing

E3 Submission

One submission per group on E3. Submission is due Mon 2026-10-19, 23:59.

Upload to E3



HW1_<TeamID>.zip



HW1_<TeamID>_report.pdf

- [ACL template](#) · At most 6 pages, excluding references.
- GitHub repository link on the first page, below the title.



HW1_<TeamID>_presentation.pptx/.pdf

- The slides you present on 2026-10-20



Rules

- One member uploads for the whole group
- Use the **file names** exactly as shown or there will be a deduction of **5 points** for HW1 paper report.
- If the **GitHub repository link** is not included on the first page of the paper report, the paper report will receive a **score of 0**.
- GitHub repository: your code, datasets and a short README
- Pages after page 6 of the paper report are not read

Mon 2026-10-19, 23:59

Tue 2026-10-20

HW1 Deadline

Report and slides on E3

HW1 Presentation

In class



No late submissions

Files uploaded after
2026-10-19 23:59
are not graded.

Grading Policy

HW1 grade = 70% report + 30% presentation + optional bonus.

70% **Report**

Scored out of 100 by section (see "Paper Structure"), then weighted 70%

30% **Presentation**

Story, evidence, clarity, timing, and Q&A

+6% **Bonus**

Up to 6% added to the HW1 grade via the LLM TA Judge (optional)

We assess whether

- Design decisions are justified
- Conclusions are supported by your own experiments
- Comparisons are fair
- Analyses cite concrete numbers and cases
- Writing and figures are clear
- The topic is creative and distinctive



Creative, distinctive topics score higher.



LLM TA Judge

Automated reviewers score your report against the published rubric and return written feedback. See the linked page for how to submit.

[How the LLM TA Judge works →](#)

Have Fun!

